WASP-3b

R-0293

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-3_180705` · created 2026-07-17T19:03:11+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2458304.7757 +/- 0.0013 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1408 +/- 0.0066
Transit depth (Rp/Rs)^2	1.98 +/- 0.19 [%]
Orbital Inclination (inc)	81.74 +/- 0.57
Airmass coefficient 1 (a1)	1.2446 +/- 0.01
Airmass coefficient 2 (a2)	-0.201 +/- 0.017
Scatter in the residuals of the lightcurve fit is	0.8 %
Best Comparison Star	None
Optimal Aperture	5.97
Optimal Annulus	12.96
Transit Duration (day)	0.1053 +/- 0.0048

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1408 ± 0.0066 vs 0.10538 (deviation 5.17σ)
Duration fit vs published	0.1053 d vs 0.1167 d (deviation 2.28σ)
Mid-time O–C	7.2 min
Depth SNR	20.5
Residual scatter	0.8 %

Light curve

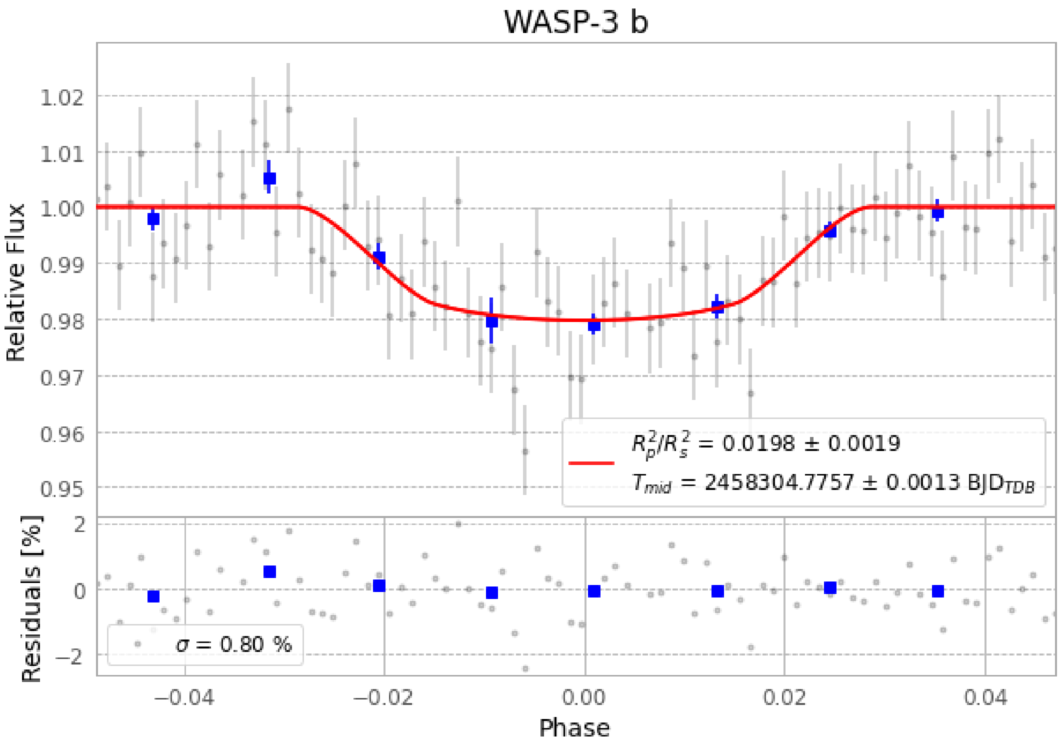

Figure 1 — FinalLightCurve_WASP-3 b_2018-07-04.png (SHA-256 0a522a64d28423e1…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [337, 198], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 3ce539840be3ba92…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a5addef`

Data provenance

86 FITS frames (56 MB), `WASP-3180705042112.FITS` → `WASP-3180705083612.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-3-180705.log (acfb0c35fd88…), FinalLightCurve_WASP-3 b_2018-07-04.png (0a522a64d284…), FinalLightCurve_WASP-3 b_2018-07-04.csv (ae2ea2fd1156…)

Compute resources

Wall time 210.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)